

TPU0S1T processed

Peak-picking modal fit after Schmitz & Smith, Machining Dynamics, 2nd ed., Sec. 2.5.1 (Fig. 2.21, Eqs. 2.58-2.63)

1. Test

File	/home/user/TPU/Transmissibility test/0%/TPU0S1T.csv
Experiment / FRF type	transmissibility (Y axis units in file: m/s^2 / m/s^2)
Conversion applied	none (transmissibility, accel/accel, dimensionless)
Sign convention	+1 (Fig. 2.21: Im < 0 at resonance)
Data	2047 points, 0.391-799.614 Hz
Search band	30-500 Hz
Smoothing	11-point quadratic Savitzky-Golay (curves and picking)
Modes fitted	1
Processed	2026-10-02 21:15

2. Picked points (Fig. 2.21)

Point	Meaning	Variable	f [Hz]	omega [rad/s]	Value [-]	Coherence
1	Im minimum, mode 1	omega_n1, A	58.203	365.70	-3.036	0.604
3	Re maximum, mode 1	omega_3	53.907	338.70	3.828	0.653
4	Re minimum, mode 1	omega_4	74.219	466.33	-1.124	0.713
2	Im minimum, mode 2	omega_n2, B	NaN	NaN	NaN	NaN

3. Modal parameters (Eqs. 2.58-2.63)

Equation	Quantity	Variables (mode 1 / mode 2)	Mode 1	Mode 2
2.58 / 2.59	damping ratio [-]	zeta_q1 / zeta_q2	0.1745	NaN
2.60 / 2.61	stiffness [-]	k_q1 / k_q2	0.9438	NaN
2.62	mass [s^2]	m_q1 / m_q2	7.057e-06	NaN
2.63	damping [s]	c_q1 / c_q2	0.0009007	NaN

Modal matrices (p. 43): [K_q] = diag(0.9438) [-] [M_q] = diag(7.057e-06) [s^2] [C_q] = diag(0.0009007) [s]

4. Notes

- Transmissibility has no force: A, B are dimensionless and k_q, m_q, c_q come out in [-], [s^2], [s], not N/m, kg, N s/m. For one mode on a moving base $Im(T) = -1/(2 \text{ zeta})$ at resonance, so Eq. 2.60 gives k_q1 close to 1.
- One mode fitted (cfg.pick_n_modes = 1): Eqs. 2.59 and 2.61 and the mode 2 parts of 2.62-2.63 are NaN.
- Coherence below 0.75 at point(s) 1, 3, 4 (0.60, 0.65, 0.71): those values are uncertain.
- Points 3 and 4 are not symmetric about point 1 (4.30 Hz below, 16.02 Hz above); for one viscous mode they lie near omega_n1(1 +/- zeta_q1).

5. Real and imaginary parts with the picked points

